

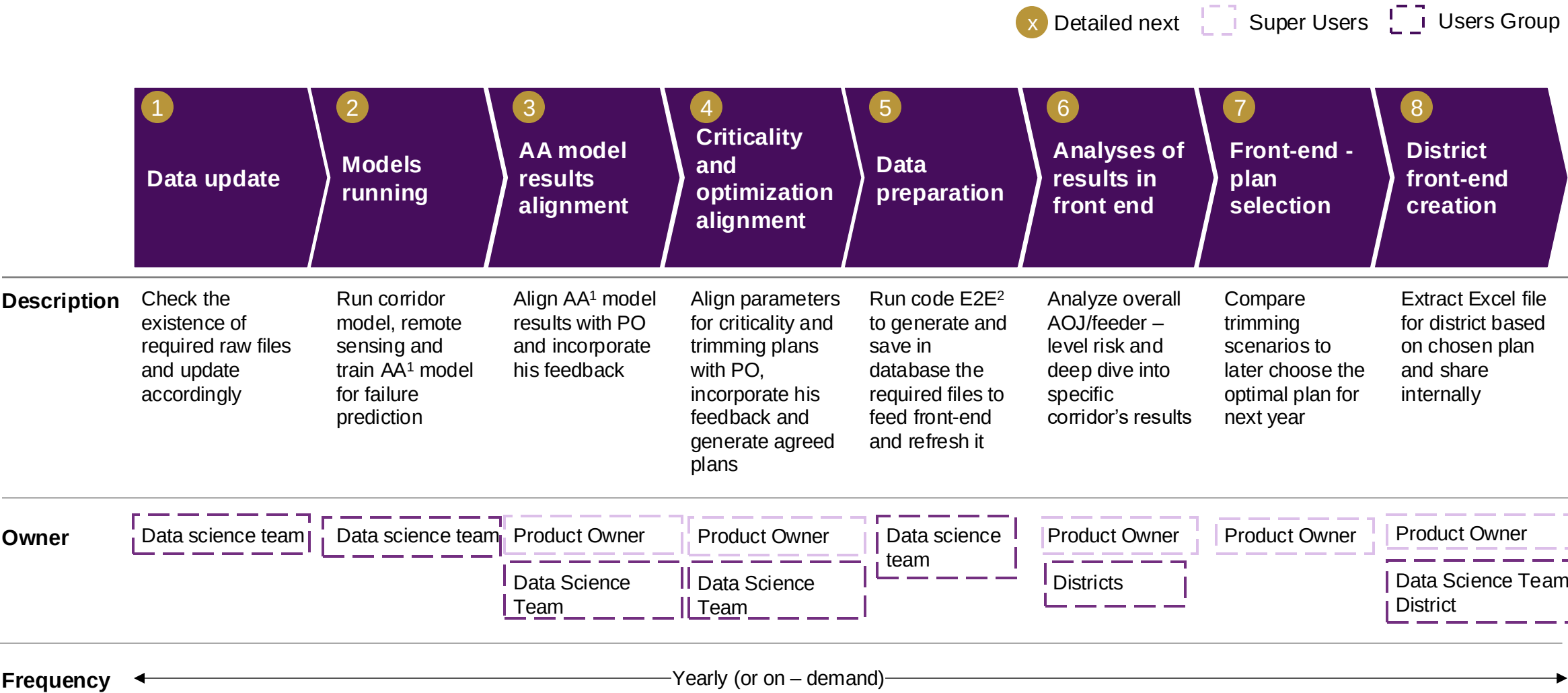


TonmaAI Vegetation Management – User Training



December 2024

Data Science team and product owner will collaborate to update TonmaAI Vegetation Management Solution to create trimming plans yearly



1. Advanced Analytics; 2. End-to-end

1 Data update – SQL structured database

Data Science team

DS will review the list of required data files in data mapping Excel...

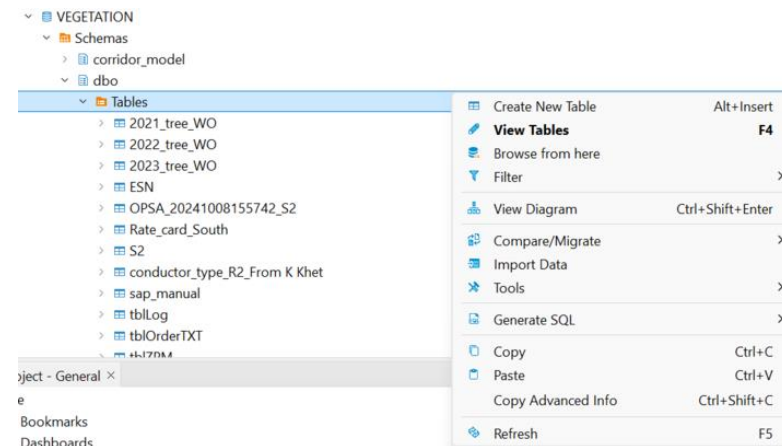
... to check for their existence and update accordingly in database

Path of each file in the database is indicated

Category	Folder/File name	Data Storage	Folder Path
outage history	1) 2023-12-01-Report No.52 Year to Date PEA-ALL_applied filter for Asset management.xlsx 2) 2022-12-01-Report No.52 Year to Date PEA-ALL.xlsx 3) 2022-01-01-รายงานที่ 52 รายปี PEA-ALL.xlsx 4) 2021-01-01-รายงานที่ 52 รายปี PEA-ALL.xlsx 5) ESN 2024.xlsx	SQL structured database	dbo
SAP trimming data	1) S2.xlsx 2) sap_manual.xlsx	SQL structured database	dbo
work order data	1) 2021 เฉพาะต้นไม้ + W.O..xlsx 2) 2022 เฉพาะต้นไม้ + W.O..xlsx 3) 2023 เฉพาะต้นไม้ + W.O..xlsx	SQL structured database	dbo

Required naming and format are defined for the code to run E2E²

Database



Files can be uploaded from database and through code

File upload through coding

```
connection_string = (
    f"mssql+pyodbc://{username}:{password}@{server}:{port}/{database}"
    "?driver=ODBC+Driver+17+for+SQL+Server&timeout=120&Encrypt=no")

engine = create_engine(connection_string)

table_name = 'F9_visualization.pbi_combined_opti_scenarios'

# Write DataFrame to SQL Server
combined_opti_scenarios.to_sql(name=table_name, con=engine, schema='final_data', if_exists='replace', index=False)
```

1 Data update – Local currently and Oracle Instance in the future

Data Science team

DS will review the list of required data files in data mapping Excel...

... to check for their existence and update accordingly in database

Category	Folder/File name	Data Storage	Folder Path
GIS data	s2_0.gdb	Local	data/corridor_model/F0_raw/extraction/
GIS data	s2_1.gdb	Local	data/corridor_model/F0_raw/extraction/
GIS data	s2_2.gdb	Local	data/corridor_model/F0_raw/extraction/
GIS data	s2_3.gdb	Local	data/corridor_model/F0_raw/extraction/
META 1m canopy detection	META	Local	data/remote_sensing/F0_raw/meta/
Sentinel-2	sentinel-2	Local	data/remote_sensing/F2_processed/

Path of each file in the database is indicated

Required naming and format are defined for the code to run E2E²

Vegetation Management > data > corridor_model > F0_raw > extraction >				
		Sort	View	...
☐	Name	Date modified	Type	Size
	S2_0.gdb	05/11/2024 8:28	File folder	
	S2_1.gdb	25/10/2024 11:08	File folder	
	S2_2.gdb	25/10/2024 11:08	File folder	
	S2_3.gdb	14/11/2024 6:30	File folder	

Vegetation Management > data > remote_sensing > F0_raw > meta >				
		Sort	View	...
☐	Name	Date modified	Type	
	imagery	26/11/2024 3:04	File folder	
	tiles	26/11/2024 23:37	File folder	

Files can be updated following the same naming and structure

2 Data processing and Advanced Analytics models running

Data Science Team

DS will run code prior to AA¹ model and adapt code as required...

... and once master data is ready, tune hyperparameters for AA¹ model...

... until optimal accuracy and engineering-wise desired key driver results are achieved

01_download_data.ipynb
02_process_data.ipynb
03_create_network_graph.ipynb
04_current_state_corridor_tracing.ipynb

Corridor model

01_filter_corridors_to_aoi.ipynb
02a_mjm_filter_to_aoi.ipynb
02b_connect_mjm_to_spans.ipynb
03_download_meta_canopy_cover.ipynb
04_clip_meta_to_aoi.ipynb
05_analyze_meta_data.ipynb
06_get_sentinel2_data.ipynb
07_remote_sensing_output.ipynb

Remote Sensing

1.0_corridor_cleaning.ipynb
1.1_pole_cleaning.ipynb
1.2_weather_cleaning.ipynb
1.3_outage_cleaning.ipynb
1.4_trimming_cleaning.ipynb
1.5_remote_sensing_cleaning.ipynb
1.6_mjm_cleaning.ipynb
2_create_model_inputs.ipynb
3_outage_probability_model.ipynb

Analytics

Code can be adapted as required in case data format has changed or other features need to be created

```
C:\Users\Francesca Andretta>python run_vegx.py
Starting pipelines: corridor_mapping, remote_sensing, analytics
Starting pipeline: run_vegx_thailand_corridor_mapping.py
```

```
### Years to be used for training, testing
### Make sure you update these and that
### The training and testing params should be
training_years_list: [2020, 2021, 2022]
testing_years_list: [2023]

# Number of parameter search trials to be
number_of_hyperparameter_search_trials: 100

### Default parameters used by the model
model_parameters_binary:
  objective: 'binary'
  metric: 'binary_logloss'
  boosting_type: 'gbdt'
  feature_pre_filter: False
  verbosity: -1
  lambda_11: 6
  lambda_12: 5
  max_depth: 100
  num_leaves: 200
  min_child_samples: 19
  min_data_in_leaf: 500
  feature_fraction: 0.18
  bagging_fraction: 0.885
  bagging_freq: 1
  force_col_wise: True

### List of independent variable names used
independent_variables:
- atmospheric_pressure_max
- atmospheric_pressure_mean
- atmospheric_pressure_min
- average_temp_mean
- corridor_length
- cum_year-2_number_point_trimming
- cum_year-2_trim
- cum_year-2_trimmed_in_season_dry_sum
- cum_year-2_trimmed_in_season_humid_sum
- cum_year-2_trimmed_km_sum
- cum_year-3_number_point_trimming
```

Years used to train the model

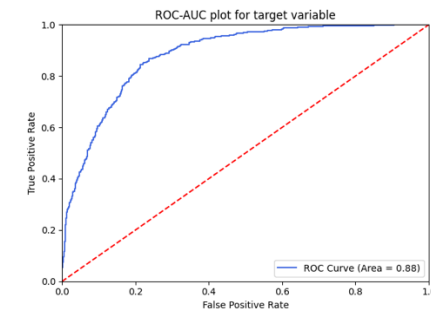
Year(s) used to test the model

of search trials for hyperparameters tuning

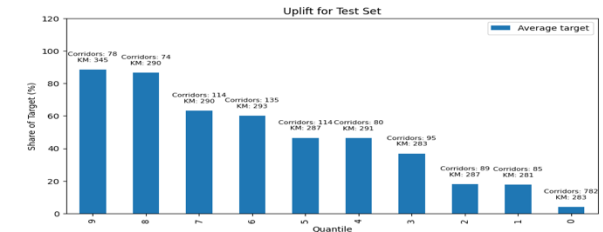
Model parameters
Target type
Evaluation metric
Boosting algorithm
Feature filter
Log level
L1 regularization
L2 regularization
Tree depth
Leaf nodes
Min child data
Min leaf data
Feature subsample
Data subsample
Sampling frequency
Column storage

Variables used in the model

ROC AUC



Deciles



Key drivers

	Feature	Importance
0	corridor_length	20.608261
1	tree_grow_in_outages_lag_1	18.610036
2	share_length_ACSR	8.985324
3	outages_lag_1	7.959630
4	km_based_density_avg	6.489368
5	share_length_PIC	5.277573
6	height_max	5.183989
7	km_based_change_density_avg	3.023925
8	density_max	2.948107
9	km_based_height_avg	2.885928

③ Results alignment for AA model

Data Science Team + Product Owner

DS and PO will meet to analyze the results of AA model so that DS can work on incorporating the feedback

Feedback from PO

A Remove features from AA¹ model

Action from DS

```
independent_variables:
- atmospheric_pressure_max
- atmospheric_pressure_mean
- atmospheric_pressure_min
- average_temp_mean
- corridor_length
- cum_year-2_number_point_trimming
- cum_year-2_trim
```

Non-desired features can easily be removed from the list of selected variables for model

B Create new features for AA model from existing data

```
training_df, testing_df, prediction_df = master_flat_file_generation.generate_master_flat_file(
    corridors_features=corridors,
    outage_features=outages,
    trimming_features=trimming,
    remote_sensing_features=remote_sensing,
    weather_features=weather,
    pole_features=poles,
    urban_rural_labels='',
    training_years_list=train_years,
    testing_years_list=test_years,
    prediction_year=outage_pred_year,
    lag_length=0,
    categorical_variables=[],
)
```

**Code for new feature
needs to be generated and
included in master table**

C Include new data source

- dbo
 - Tables
 - 2021_tree_WO
 - 2022_tree_WO
 - 2023_tree_WO
 - ESN

New data source needs to be uploaded into database

- 1.0_corridor_cleaning.ipynb
- 1.1_pole_cleaning.ipynb
- 1.2_weather_cleaning.ipynb
- 1.3_outage_cleaning.ipynb
- 1.4_trimming_cleaning.ipynb
- 1.5_remote_sensing_cleaning.ipynb

Data needs to be processed following same steps and included in master data

C Analyses on specific features

2. SAP data analysis indicate that outliers happen in areas trimmed recently, demonstrating trimming tends to be concentrated in risky places and may not be efficient enough in some cases

Correlates with recurred trimming rate for some areas **and in the first months after the trimming activity**

1. We noticed that **200 SAP trimming work** was **mostly concentrated in areas with high population** (see table for a full breakdown in the trimming path)

2. **10% of the corridors with good trimming had an average length**, against 10% of corridors without recurred trimming had an average length

3. For outliers in corridors with trimming in the past 12 months, **the average length of corridors was measured on average 122 days** (i.e. 4 months) after the trimming activity

Distribution of trimming activities before outliers

Trimming Activity	Outliers	Non-outliers
Trimmed in the last 12 months	4,318	432
Trimmed in the last 12 months or earlier	2	1,077
Total	4,320	1,509

40% of outliers with trimming in the last 12 months were outliers in the last 12 months after the trimming activity

Source: SAP data, 2018

2. Corridors trimmed frequently shows a fast increase of outage risk in the first 6 months

Percentage of corridors with outages in 10 months since trim¹ for corridors trimmed frequently

Months since trim	% of corridors experiencing an outage
01	12
02	18
03	20
04	20
05	20
06	20
07	10
08	15
09	10
10	15

Outage percentage increases by approximately 40% (from 12% to 16%) over the first six months following a trim, with an average monthly increase of 6.66% during this period.

1. Data points are based on 2014 corridor outages, representing the highest number of outages in any year since the corridor was last trimmed.

- Corridors trimmed frequently:
 - Are in high-risk areas, as the risk of outage increases rapidly during the initial months following trimming
 - Have the highest risk of experiencing outages in the first months after being trimmed, leading them to fail multiple times

Ad-hoc notebooks for analyses

4 Results alignment for criticality and optimization

Data Science Team + Product Owner

DS and PO
will meet to
align
parameters
for criticality
and
optimization
so that DS
can work on
incorporating
the feedback

Feedback from PO

A Adapt criticality calculation

Action from DS

Create criticality score

```
criticality_df,_ = criticality.calculate_criticality(corridors, wo_concat, avg_durations, criticality_customer_parameters, criticality_load_parameters, criticality_params['criticality_choice'])
```

B Adapt existing trimming frequency options

```
optimization_parameters:
degradation_time_factor: 0.5 # 6 months
## Max percentage of risk reduction by trimming
risk_removed_factor: 0.4 # = 40%
## Possible frequencies (in years)
# If it is possible to assume that the minimum frequency is zero
# which means that some corridors are never trimmed
# put a very high number (eg 1000) to T1
T1: 1 # 1 year
T2: 1 # 1 year
T3: 0.5 # 6 months
T4: 0.333 # 4 months
T5: 0.333 # 4 months
## Optimization level
item_name: 'corridor' # line corridor
## Asset ID
id_col: 'nearest_upstream_device' #line_global_id nearest_upstream_device
```

Parameters can be easily adapted in the code

Time after which the risk reached each maximum

Maximum percentage of risk removed by trimming, estimated through FMEA

The code is flexible to accept up to 5 frequency options. Once the PO selects the operationally viable options, the DS team can modify the parameters. If less than 5 frequencies are chosen, the DS will assign the same frequency to multiple T parameters; the code will select only the unique frequencies (in this case 1/year, 2/year, 3/year)

C Adapt optimization parameters (e.g., risk removed, degradation time)

D Generate trimming plans given max budget or SAIFI objective

```
methodology_actions = methodology_actions
max_risk_reduction_perc_actions = max_risk_reduction_perc_actions_chosen
actions_calendar=None
prioritized_actions = optimization.get_action_parameters(actions_inputs=actions_inputs_keep)
```

```
C:\Users\Francesca Andretta\vegx_thailand_analytics>python run_vegx_analytics.py --chosen_budget 13_693_167
Pipeline execution started.
Running notebooks/3_outage_probability_model.ipynb...
```

5 Data preparation

Data Science Team

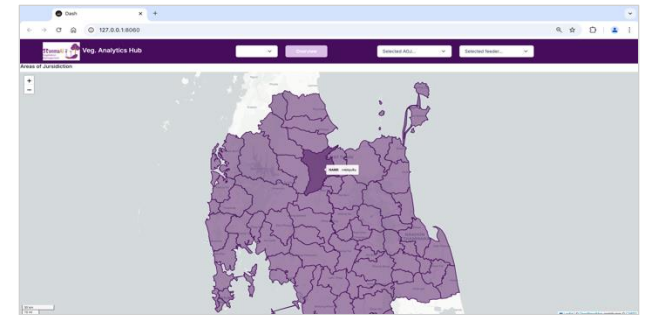
Once AA¹ model, criticality and optimization are aligned with PO, DS will run code E2E²...

```
C:\Users\Francesca Andretta>python run_vegx.py --chosen_risk_reduction 0.25
Starting pipelines: corridor_mapping, remote_sensing, analytics
Starting pipeline: vegx_thailand_corridor_mapping/run_vegx_thailand_corridor_mapping.py
Pipeline execution started.
Running notebooks/01_download_data.ipynb...
```

to generate and save in database the data required to feed front-end...

- ▼ risk_modeling
 - ▼ Tables
 - > F7_front_end_inputs.front_end_input_features
 - > F7_front_end_inputs.model_shap_features
 - > F7_front_end_inputs.optimizations_metrics
 - > F7 front end inputs.optimizations scenarios

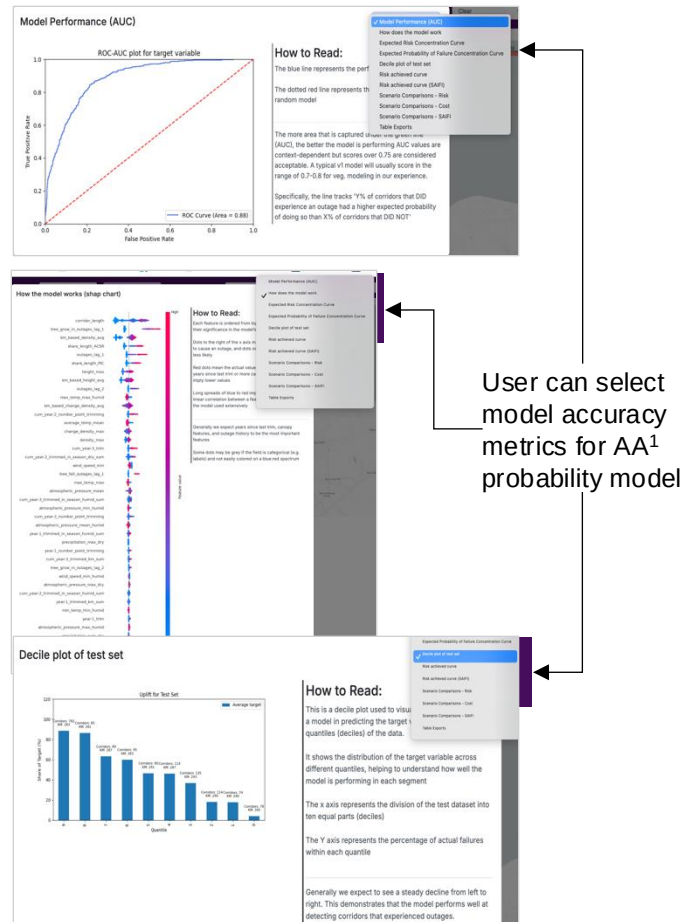
... and make sure the data is refreshed in Dash - <https://dev-tonmai-tcc.pea.co.th/>



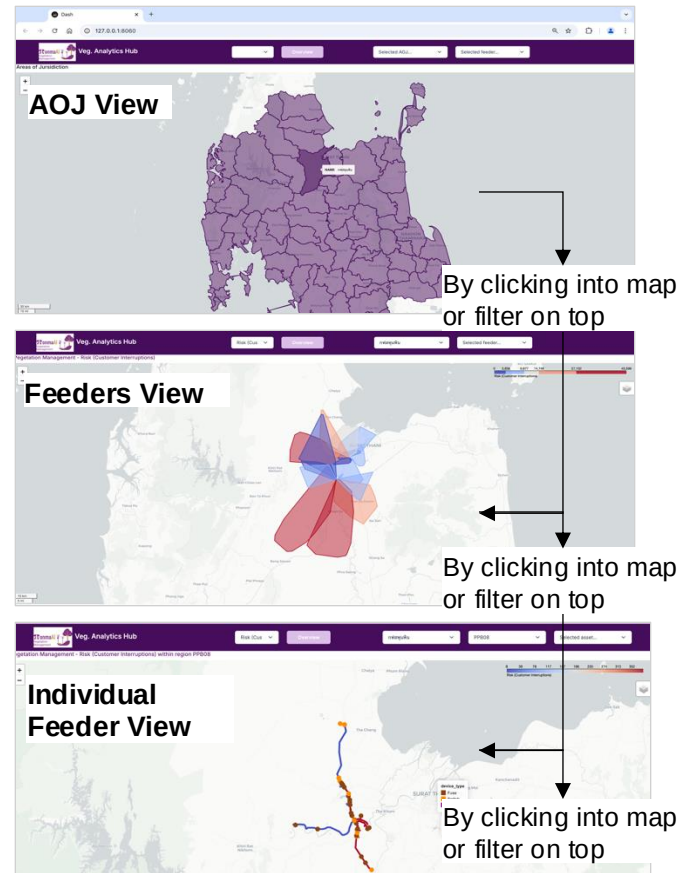
6 Analyses of results in front-end - <https://dev-tonmai-tcc.pea.co.th/>

Product Owner and district

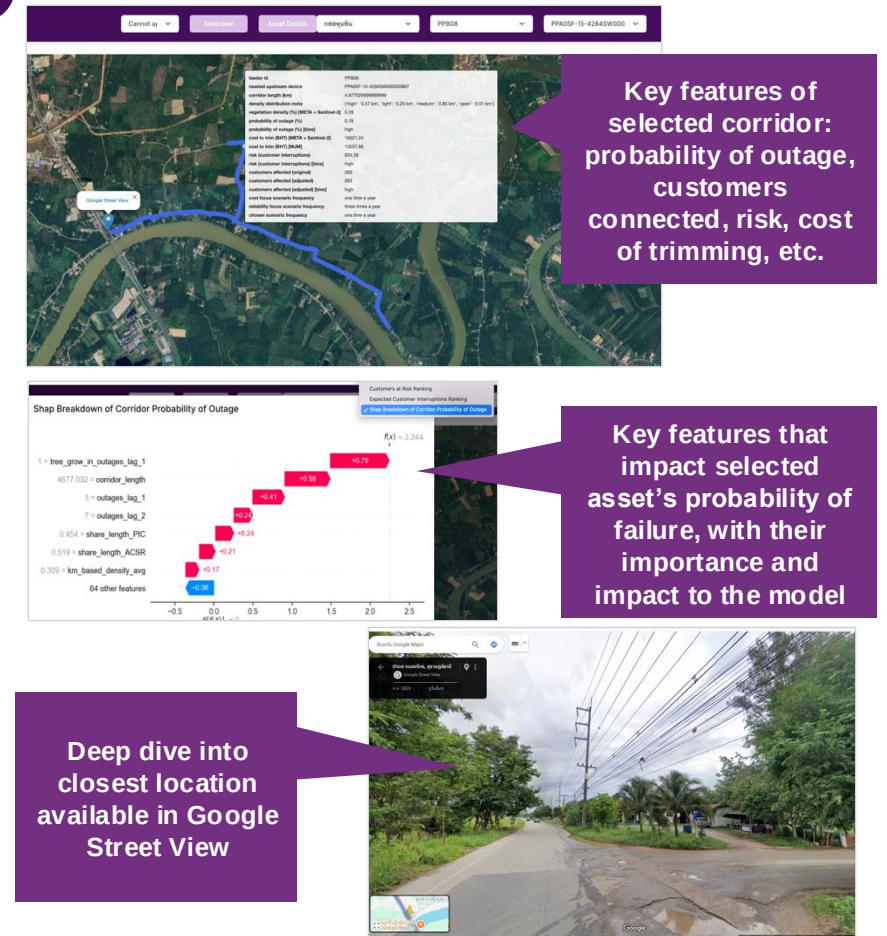
User will analyze AA¹ model accuracy...



... then move into overall AOJ or feeder - level risk, probability and consequence of failure...



... to later select a specific corridor and deep dive into its details

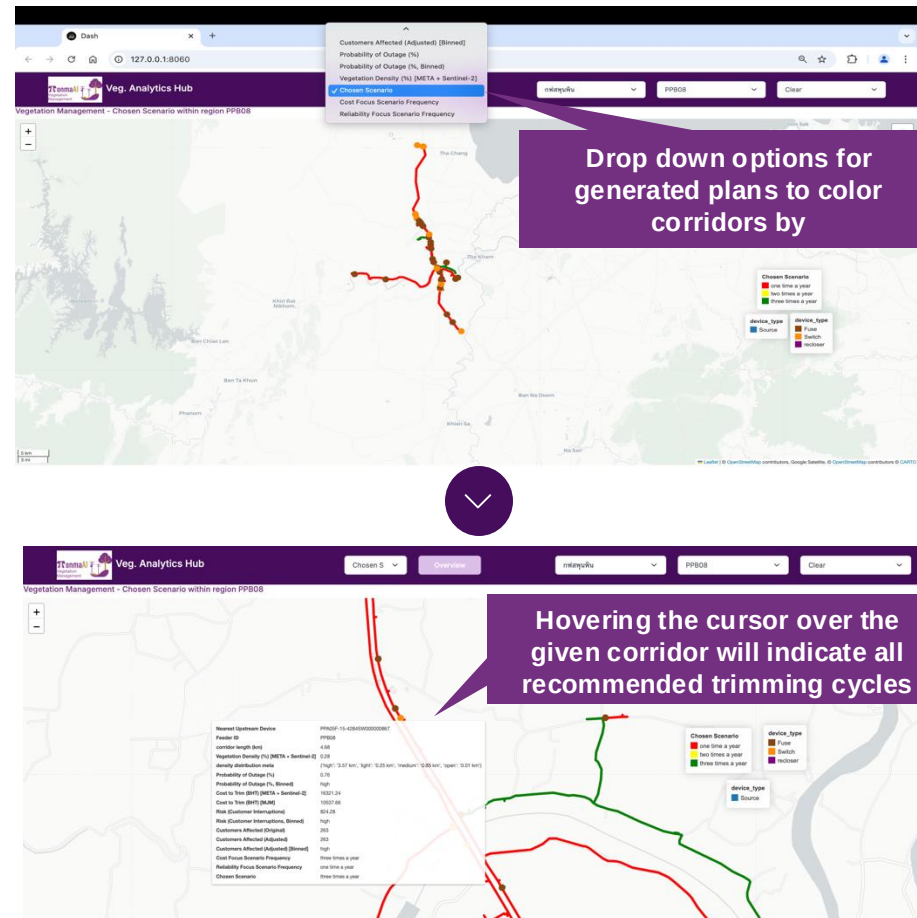
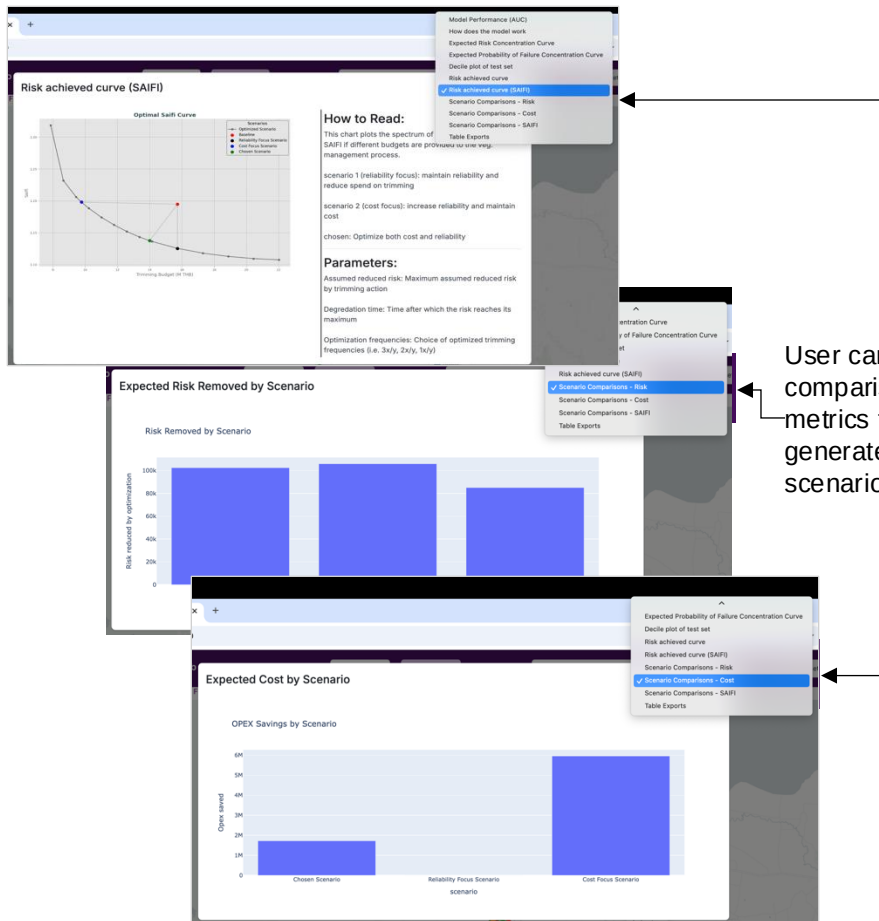


7 Trimming plan selection in front-end

Product Owner

User can compare already generated scenarios in terms of risk achieved, saved cost, SAIFI...

... as well as analyze the suggested trimming frequency for each plan and corridor



Product Owner will choose the optimal trimming plan for next year

8 Plan extraction for district

Product Owner + Districts

Once Product Owner has chosen the optimal trimming plan for next year

Product Owner will extract the file containing information at corridor level...

Asset overview

feeder id	nearest upstream device	nearest upstream device list	corridor length (km)
PPA04	BOA0SW-01-4284RC000000004	['42SWKA000044732', '42SWKA000015942', '42SWKA000011724', '42SWKA000009897', '42SWKA000009894', '42RCCKA000010602', '42SWKA000025782', '42SWKA000115701', '42SWKA000061608', '4284SW000000180', '4284SW000000059', '42SWKA000092224', '42SWKA000051411', '42SWKA000064001', '42SWKA000043350', '42SWKA000028102']	3.69423

... to share chosen plan with districts for plan implementation

region	feeder	nearest upstream device	upstrs	425SWKA	geomc	for len	distribi	of outa	of outa	of outa	of outa	ility of	of out	hmer le	br inte	ffect	ected	scenari	quency	
phun_phin	PPA03	PPA04F-11-03-4284SW0000000496	4284SW0	[42SWKA MULTILINI	0,45856	'high': 0	0	0	0	0	0,05	low	4,04	medium	110,47	high	one time a year			
phun_phin	PPA05	PPA05F-11-4284SW0000000652	4284SW0	None	MULTILINI	0,48276	'light': 0	0	0	0	0,04	low	2,16	medium	177	high	one time a year			
phun_phin	PPA08	PPA08F-873-425WKA000103271	425WKA0	[425WKA MULTILINI	1,01992	'high': 0	0	0	0	0	0,11	medium	7,85	medium	92,01	high	one time a year			
phun_phin	PPA08	PPA08F-93-425WKA000110349	425WKA0	None	MULTILINI	0,50166	'medium'	0	0	0	0,04	low	0,6	low	49,54	medium	one time a year			
phun_phin	PPA08	PPA03F-230-4284SW0000000724	4284SW0	None	MULTILINI	0,28622	'high': 0	0	0	0	0,04	low	1,97	medium	161,78	high	one time a year			
phun_phin	PPA02	PPA02F-02-4284SW0000000861	4284SW0	[425WKA MULTILINI	1,20245	'high': 0	1	2	4	5	0,37	high	94,98	high	118	high	two times a year			
phun_phin	PPB07	PPB07F-18-4284SW00000003513	4284SW0	[425WKA MULTILINI	1,29561	'high': 1	3	5	5	7	0,45	high	147,65	high	129	high	two times a year			
phun_phin	PPB03	PPB03F-76-4284SW00000003417	4284SW0	None	MULTILINI	0,47772	'light': 0	0	0	3	5	0,13	medium	20,29	medium	104	high	two times a year		
phun_phin	PPA02	PPA02F-11-4284SW0000000609	4284SW0	None	MULTILINI	0,6021	'light': 0	1	2	2	2	0,33	high	62,76	high	83	medium	two times a year		
phun_phin	PPA09	PPA09F-37-4284SW0000001069	4284SW0	[425WKA MULTILINI	3,35986	'high': 0	1	3	4	9	0,47	high	204,67	high	178,82	high	two times a year			
phun_phin	PPA01	PPA01WR-05-4284SW00000013222	4284SW0	[425WKA MULTILINI	0,99443	'high': 0	0	0	0	0	0,17	medium	201,9	high	689,83	high	three times a year			
phun_phin	PPB03	PPB03F-53-4284SW00000003446	4284SW0	None	MULTILINI	0,9369	'high': 0	10	21	28	36	0,72	high	143,9	high	45,91	medium	three times a year		
phun_phin	PPB06	PPB06F-18/1-4284SW00000003517	4284SW0	None	MULTILINI	1,38126	'high': 1	7	9	9	10	0,71	high	184,91	high	59	medium	three times a year		
phun_phin	PPA01	PPA06F-17-4284SW0000000872	4284SW0	[425WKA MULTILINI	0,99911	'high': 0	0	0	0	0	0,25	medium	271,88	high	683,83	high	three times a year			
phun_phin	PPA01	PPA01F-62-425WKA000045268	425WKA0	None	MULTILINI	1,22035	'high': 1	2	2	3	8	0,75	high	274,19	high	87,48	medium	three times a year		
phun_phin	PPA04	BOA0SW-01-4284RC0000000004	4284RC0C	[425WKA MULTILINI	3,69423	'high': 0	0	2	3	5	0,13	medium	342,47	high	4011,78	high	three times a year			
phun_phin	PPA02	PPA02WR-01-4284RC0000000007	4284RC0C	[4284SW MULTILINI	11,5677	'high': 3	0	0	0	0	0,17	medium	996,93	high	3406,19	high	two times a year			
phun_phin	PPA09	PPA09WR-01-4284RC0000000008	4284RC0C	[425WKA MULTILINI	8,82898	'high': 0	0	0	0	0	0,19	medium	2931	high	10014,3	high	three times a year			
phun_phin	PPA09	PPA09WR-02-4284RC0000000009	4284RC0C	[4284RC MULTILINI	12,5177	'high': 1	0	1	2	3	0,16	medium	1631,27	high	8360,26	high	three times a year			
phun_phin	PPA01	PPA01WR-03-4284RC0000000010	4284RC0C	[4284SW MULTILINI	9,5318	'high': 3	3	4	6	10	0,78	high	2342,5	high	7470,12	high	three times a year			
phun_phin	PPB03	PPB03WR-02-4284RC0000000011	4284RC0C	[4284RC MULTILINI	12,6623	'high': 4	0	0	0	3	0,16	medium	1880,89	high	9639,54	high	three times a year			
phun_phin	PPA01	PPA01WR-01-4284RC0000000012	4284RC0C	[4284SW MULTILINI	18,5415	'high': 0	1	1	1	2	0,23	medium	1042,08	high	2620,99	high	two times a year			
phun_phin	PPB06	PPB06WR-03-4284RC0000000062	4284RC0C	[425WKA MULTILINI	23,2495	'high': 0	1	1	7	20	0,22	medium	3627,32	high	9123,26	high	three times a year			
phun_phin	PPB03	PPB03WR-03-4284RC0000000063	4284RC0C	[4284RC MULTILINI	22,0006	'high': 3	1	2	12	23	0,29	high	4619,28	high	6886,92	high	three times a year			

Only chosen plan column should be kept in file before sharing with district

